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- **Spontaneous Parametric Down-Conversion (SPDC)** is a key method for generating entangled photon pairs for quantum communication, computation, and sensing.
- **Type-II SPDC:** a pump photon splits into orthogonally polarized signal and idler photons inside a χ^2 nonlinear crystal.
- The **photon pair rate and heralding efficiency** depend strongly on pump focusing, beam waists, crystal length, and phase-matching temperature.
- **PPKTP** is preferred due to its high nonlinear coefficient, stability, and wide temperature-tuning range.
- Precise modelling helps predict **optimal focusing conditions** for maximizing pair generation.
- Understanding these dependencies is crucial for building **high-brightness, low-loss quantum light sources**.

Our approach predicts a QPM temperature of **40.66 °C** for degenerate Type-II SPDC in PPKTP. The optimal pump waist of **119.22 μm** yields a maximum brightness of **2595.60 pairs/s.mW**, with beam-waist matching playing the most significant role in enhancing pair generation. Crystal length and pump power further increase the rate but with diminishing returns due to spatial-overlap limits.

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